Mihir Kavishwar

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Research Interests

Mixed-Signal VLSI Design, In-Memory Computing, High-Speed Links, Massive MIMO, Edge AI

Education

University of Illinois Urbana-Champaign, Illinois, USA [2023 - Present]
- **PhD candidate**, Department of Electrical and Computer Engineering, **GPA**: 4.00 / 4.00
- **Advisor**: Prof. Naresh Shanbhag

Princeton University, New Jersey, USA (transferred out) [2022 - 2023]
- **PhD student**, Department of Electrical and Computer Engineering, **GPA**: 3.55 / 4.00

Indian Institute of Technology Bombay, Mumbai, India [2017 - 2022]
- **Dual Degree** (B.Tech + M.Tech) in Electrical Engineering, **GPA**: 9.35 / 10.00
- **Specialization**: Microelectronics, **Minor**: Systems and Control Engineering

Publications

1. (submitted) R. Y. Fu*, **M. Kavishwar***, G. Lee, HM. Ou, O. Bekdache and N. R. Shanbhag, "A 2.14 pJ/b, 76.6 Gb/s/mm^2 SRAM-Based Analog In-Memory Detector for mmWave Massive MIMO," 2027 IEEE International Solid State Circuits Conference (ISSCC).
2. **M. Kavishwar** and N. R. Shanbhag, "Minimizing ADC Precision for Analog In-Memory Computing," 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Barcelona, Spain, 2026, pp. 20446-20450, doi: 10.1109/ICASSP55912.2026.11465137.
3. R. Y. Fu, **M. Kavishwar** and N. Shanbhag, "Detector-in-Memory for Massive MIMO Detection," IEEE International Symposium on Circuits and Systems (ISCAS), Shanghai, China, 2026, pp. 1122-1126, doi: 10.1109/ISCAS66217.2026.11562752.
4. **M. Kavishwar** and N. Shanbhag, "Energy-Accuracy Trade-Offs in Massive MIMO Signal Detection Using SRAM-Based In-Memory Computing," in IEEE Transactions on Circuits and Systems I: Regular Papers, doi: 10.1109/TCSI.2025.3594230, 2025.
5. **M. Kavishwar** and N. Shanbhag, "Massive MIMO Signal Detection using SRAM-based In-Memory Computing," IEEE International Symposium on Circuits and Systems (ISCAS), Singapore, pp. 1-5, doi: 10.1109/ISCAS58744.2024.10558118, 2024.
6. **M. Kavishwar**, P. Kurrey and R. Zele, "Analog Acoustic Feature Extraction and Delta-Sigma Modulation-based Neural Network Classification for Voice Activity Detection," IEEE India Council International Conference (INDICON), Kochi, India, pp. 1-4, doi: 10.1109/INDICON56171.2022.10039873, 2022.
7. P. Kurrey, **M. Kavishwar** and R. Zele, "Analog/Mixed-Signal Classification for Voice Activity Detection," IEEE International Conference on Electronics, Circuits and Systems (ICECS), Glasgow, United Kingdom, pp. 1-4, doi: 10.1109/ICECS202256217.2022.9971057, 2022.

Scholastic Achievements

- Received **Paul D. Coleman Outstanding Research Award** at University of Illinois [2026]
- Awarded **Gordon Wu Fellowship** for pursuing doctoral studies at Princeton University [2022]
- Received **Undergraduate Research Award** for exceptional work in the Dual Degree Project [2022]
- Awarded **Excellence in Teaching Assistantship** in Digital Systems course [2021]
- Secured **All India Rank 287** in IIT-JEE Advanced among 160,000+ candidates [2017]

Technical Skills

Tapeout: Led digital design of detector-in-memory chip taped-out in TSMC 28nm CMOS [October'26]
Design Tools: Cadence Virtuoso, Innovus, Synopsys Verdi, Xilinx Vivado, KiCAD, KLayout, GTKWave
Programming: C++, Python, MATLAB, SPICE, Verilog, Verilog-AMS, SystemVerilog, VHDL

Key Courses

Devices: Solid State Devices, Physics of MOS Transistors, VLSI Technology
Circuits: Analog, RF, Digital, Mixed-Signal, Delta-Sigma, SerDes, Neuromorphic, Photonics
Systems: Computer Architecture, Computer Networks, Systems for ML, VLSI CAD
Algorithms: Digital Signal Processing , Data Structures and Algorithms, Machine Learning
Miscellaneous: Random Processes, Digital Communications, State Estimation, Optimization

Industry Experience

Modeling and Verification of Optical PHY in High-Speed Interconnects [June'26 - Aug'26]
NVIDIA | Manager: Wei Yu
- Investigated digital control algorithms for polarization control in high-speed optical interconnects
- Performed modeling, simulation, and verification of mixed-signal control circuits for optical PHY applications

System Design of PCI Express 6.0 Receiver [May'21 - July'21]
Ceremorphic India Pvt. Ltd. | Manager: Prof. Nijwm Wary
- Proposed system architecture for PCIe6 receiver to compensate 33 dB channel loss at 16 GHz in PAM-4
- Generated an IBIS-AMI model using MATLAB SerDes toolbox and performed transient simulations
- Achieved 3.46 dB channel operating margin with statistical simulations of proposed system architecture

Verification of FPGA-based High Frequency Trading Accelerator [May'20 - July'20]
A.P.T. Portfolio Pvt. Ltd. | Manager: Vivek Panikkar
- Completed training in universal verification methodology with SystemVerilog and cocotb
- Emulated MicroBlaze processor in QEMU and proposed an efficient method for verifying bare-metal code

Design and Analysis of Receiver Equalizers in High-Speed Interconnects [May'19 - July'19]
STMicroelectronics N.V. | Manager: Paras Garg
- Designed CTLE and 3-tap adaptive DFE in 32nm LP CMOS for an LVDS receiver operating at 800 Mbps
- Designed a StrongArm latch with 30 mV sensitivity to function as decision slicer at the receiver
- Successfully reduced inter-symbol interference and achieved 500 mV eye height post equalization

Positions of Responsibility

Technical Session Chair, CSL Student Conference [Aug'25 - Present]
UIUC | Coordinated Science Laboratory Student Conference is a student-organized annual event of UIUC
- Co-chaired the Circuits, Architectures, and Scalable Systems session, facilitating discussions for 40+ attendees
- Led speaker outreach and coordination; organized and moderated oral and poster presentations

CUbiC Task Leader, PI: Prof. Shanbhag [Dec'23 - Present]
UIUC | Center for Ubiquitous Connectivity is one of the seven JUMP 2.0 research centers
- Nominated as the task leader for Programmable Energy-Efficient DSP Architectures task
- Responsible for coordinating efforts and ensuring effective communication with industry liaisons

Graduate Teaching Assistant [Jan'22 - May'22]
IIT Bombay | Course: Mixed Signal VLSI Design | Instructor: Prof. Rajesh Zele
- Assisted instructor in managing logistics and course plan for a class of 70+ students
- Responsible for ideation, supervision and evaluation of course project on design of DAC with Cadence tools

Undergraduate Teaching Assistant [July'21 - Nov'21]

IIT Bombay | *Course: Digital Systems* | *Instructor: Prof. Siddharth Tallur*

- Assisted professor in managing logistics and course plan for a class of 200+ students
- Designed assignments and examinations, conducted tutorials and evaluated answer scripts

Editorial Board Member, BackgroundHum [May'21 - May'22]

IIT Bombay | *BackgroundHum is the official student newsletter of the Department of Electrical Engineering*

- Lead panelist for articles on the global chip shortage and demystifying the PhD program

Institute and Department Academic Mentor [July'20 - July'21]

IIT Bombay | *Student Mentorship Program enables mentorship of junior students by their seniors*

- Mentored 14 undergradute freshmen by providing guidance in academic and extracurricular endeavors
- Mentored 6 sophomores in the EE department with the focus on helping academically weaker students